

Boqi (Percy) Chen

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Education

2021/01–2025
Montreal, Canada

- 📖 **Doctor of Philosophy (PhD) – McGill University** in Software Engineering
 - Supervisors: **Gunter Mussbacher** and **Dániel Varró**.
 - Thesis title: *Domain-Driven and Consistent Integration of Large Language Models*.
 - Expected defense date: October 2025.**

2016/09–2020/12
Montreal, Canada

- 📖 **Bachelor of Software Engineering – McGill University**
 - GPA: **3.99/4.0** (Dean's Honour List).

Research Experience

2025/05 – 2025/06
Halifax, Canada

- 📖 **Visiting Researcher – Dalhousie University**
 - Visiting the Software Maintenance and Analytics Research Team (SMART) at the Faculty of Computer Science.
 - Supervisor: **Tushar Sharma**
 - Topic: *Non-functional Evaluation of Large Language Models for Software Design*

2023/10
Budapest, Hungary

- 📖 **Visiting Researcher – Budapest University of Technology and Economics**
 - Visiting the Critical Systems Research Group (FTRSG) at the Department of Artificial Intelligence and Systems Engineering.
 - Supervisor: **Oszkár Semeráth**
 - Topic: *Certifying the Robustness of Graph Neural Networks using Abstract Interpretation Techniques*

Industrial Experience

2021/09 – present
Toronto, Canada
(part-time)

- 📖 **Machine Learning Engineer – Aggregate Intellect**
 - Collaboration through a Mitacs Accelerate grant.
 - Supervisor: **Amir Feizpour**
 - Maintaining the open-source library *Sherpa AI* for orchestrating LLM agents with software models.
 - Designed a state machine-based framework for integrating domain-specific practices into LLMs.
 - Led the end-to-end development of multiple customized LLM-based applications addressing diverse client requirements.

2024/05 – 2025/04
Toronto, Canada
(part-time)

- 📖 **Research Associate Intern – Huawei Canada**
 - Research group: Intelligent Testing Lab at Huawei Waterloo Research Center.
 - Supervisors: **Ou Wei** and **Bingzhou Zheng**
 - Enhanced an LLM-based pipeline that transforms natural language requirements into testing models, substantially increasing production acceptance rates by over 20%.
 - Proposed and implemented a novel approach to provide guarantees on the consistency of LLM-generated models with domain-specific constraints.
 - Contributed to an internal tool to automatically evaluate the quality of LLM-generated models.

2018/05 – 2018/08
Montreal, Canada

■ **Game Developer Intern – Behaviour Interactive**

- Participated in the development and maintenance of a mobile game in Unity.
- Solved multiple platform-dependent bugs for iOS and Android legacy bugs.
- Worked on the design and implementation of new game levels.

Teaching and Supervision

Teaching

2021/01–2024/12
Montreal, Canada

■ **Teaching Assistant – McGill University**

- **ECSE 223 Model-Based Programming** (undergraduate, 8 times).
- **ECSE 439/539 Software Language Engineering** (graduate, 2 times).
- **ECSE 326 Software Requirements Engineering** (undergraduate, 1 time).
- **ECSE 429 Software Validation** (undergraduate, 2 times).
- Instructors: Gunter Mussbacher, Dániel Varró and Marwan Kanaan.
- Duties: Teaching tutorials, grading assignments and examples, holding office hours, and preparing course project materials.

2022/11
Atlanta, Georgia
(Online)

■ **Guest Lecturer – Georgia State University**

- Course title: **Advanced Deep Learning Analytics**.
- Instructor: Kulsoom Abdullah.
- Talk title: Graph Neural Networks and Their Applications.

2017/09–2020/12
Montreal, Canada

■ **Course Mentor – McGill University**

- **ECSE 223 Model-Based Programming** (undergraduate, 2 times).
- **ECSE 321 Intro to Software Engineering** (undergraduate, 1 time).
- **MATH 140 Calculus 1** (undergraduate, 1 time).
- Instructors: Gunter Mussbacher, Dániel Varró and Sidney Trudeau.
- Duties: Holding mentor sessions, answering questions, grading assignments, and developing an automated grading framework for course assignments.

Supervision

2024/09 – present
Montreal, Canada

■ **Undergraduate Student Co-supervisor – McGill University**

- Co-supervising an undergraduate student in a research project.
- Project topic: *Improving LLM Code Generation with Model-Driven Techniques*.

2024/09 – 2025/04
Montreal, Canada

■ **Design Project Co-supervisor – McGill University**

- Co-supervised a team of 4 undergraduate students in a design project.
- Project topic: *Structured State Machine Modeling with Large Language Models*.

2022/09 – 2024/09
Montreal, Canada

■ **Master Student Mentor – McGill University**

- Co-supervised 2 master students in their research projects and master's theses.
- Project topic: *Automated Generation and Evaluation of Domain Models with Large Language Models*.

Academic Services

Organizing Committee Member

2025 ➤ **Model-Driven Requirements Engineering Workshop (MoDRE)**

Program Committee Member

2025 ➤ **International Workshop on Multi-Agent Systems using Generative Artificial Intelligence for Automated Software Engineering (MAS-GAIN)**

➤ **Large Language Models for Software Engineering Workshop (LLM4SE)**

- 2024 > **Int'l Conference on Model Driven Engineering Languages and Systems (MODELS)**
Poster track
- > **Large Language Models for Model-Driven Engineering Workshop (LLM4MDE)**

Reviewer

- 2025 > **The Journal of Systems and Software (JSS)**
- > **Expert Systems with Applications**
- > **ACM Transactions on Software Engineering and Methodology (TOSEM)**
- 2024, 2025 > **Information and Software Technology (IST)**
- 2024 > **Software and Systems Modeling (SoSyM)**
- 2023 > **Int'l Conference on Fundamental Approaches to Software Engineering (FASE)**
Subreviewer

Student Volunteer

- 2022 > **Int'l Conference on Model Driven Engineering Languages and Systems (MODELS)**

Awards

- 2025 > **Bronze Medal in the ACM Student Research Competition** at the International Conference on Software Engineering (ICSE)
- > **Graduate Research Enhancement and Travel Award** from McGill University
- 2023 – 2025 > **Accelerate Award** from Mitacs
- 2022 – 2025 > **FRQNT B2X Doctoral Scholarship** from Fonds de Recherche du Québec
- 2021 – 2024 > **McGill Engineering Doctoral Award (MEDA)** from McGill University
- 2019 > **Summer Undergraduate Research in Engineering Award** from McGill University
- 2018 – 2020 > **McGill Engineering Faculty Award** from McGill University
- 2017 – 2020 > **Dean's Honour List** from McGill University

Research Publications

Journal Articles

- 1 J. A. H. López, **B. Chen**, M. Saad, T. Sharma, and D. Varró, "On inter-dataset code duplication and data leakage in large language models," *IEEE Transactions on Software Engineering*, vol. 51, no. 1, pp. 192–205, 2025.
- 2 F. Khan, **B. Chen**, D. Varró, and S. McIntosh, "An empirical study of type-related defects in python projects," *IEEE Transactions on Software Engineering*, vol. 48, no. 8, pp. 3145–3158, 2021.
- 3 O. Semeráth, A. A. Babikian, **B. Chen**, *et al.*, "Automated generation of consistent, diverse and structurally realistic graph models," *Software and Systems Modeling*, vol. 20, no. 5, pp. 1713–1734, 2021.

Conference Proceedings

- 1 K. Ahmed, J. Song, **B. Chen**, O. Wei, and B. Zheng, "MCeT: Behavioral model correctness evaluation using large language models," in *ACM / IEEE 28th International Conference on Model Driven Engineering Languages and Systems (MODELS)*, 2025, Grand Rapids, USA, October 5-10, 2025, IEEE, 2025, (to be published).

- 2 A. A. Babikian, **B. Chen**, and G. Mussbacher, "Exploring large language models for requirements on string values," in *2nd Workshop on Multi-disciplinary, Open, and integRatEd Requirements Engineering (MO2RE 2025)*, Ottawa, Canada, April 2025., IEEE, 2025.
- 3 **B. Chen**, "Consistent graph model generation with large language models," in *47th IEEE/ACM International Conference on Software Engineering: ICSE 2025 Companion Proceedings*, Ottawa, Canada, April 27-May 3, 2025, IEEE, 2025.
- 4 **B. Chen**, A. A. Babikian, *S. Feng, D. Varró, and G. Mussbacher, "LLM-based requirements verification through consistent text data and checker generation," in *33rd IEEE International Requirements Engineering Conference, RE 2025, Valencia, Spain, September 1-5, 2025*, IEEE, 2025, (to be published).
- 5 **B. Chen**, *K. Chen, J. A. H. López, G. Mussbacher, D. Varró, and A. Feizpour, "SHERPA: A model-driven framework for large language model execution," in *ACM / IEEE 28th International Conference on Model Driven Engineering Languages and Systems (MODELS)*, 2025, Grand Rapids, USA, October 5-10, 2025, IEEE, 2025, (to be published).
- 6 **B. Chen**, J. A. H. López, G. Mussbacher, and D. Varró, "The power of types: Exploring the impact of type checking on neural bug detection in dynamically typed languages," in *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*, Ottawa, Canada, April 27-May 3, 2025, IEEE, 2025, pp. 625–625.
- 7 **B. Chen**, O. Wei, B. Zheng, and G. Mussbacher, "Accurate and consistent graph model generation from text with large language models," in *ACM / IEEE 28th International Conference on Model Driven Engineering Languages and Systems (MODELS)*, 2025, Grand Rapids, USA, October 5-10, 2025, IEEE, 2025, (to be published).
- 8 M. Saad, J. A. H. López, **B. Chen**, D. Varró, and T. Sharma, "An adaptive language-agnostic pruning method for greener language models for code," in *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering, FSE 2025, Trondheim, Norway, June 23-27, 2025*, ACM, 2025.
- 9 *K. Chen, **B. Chen**, *Y. Yang, G. Mussbacher, and D. Varró, "Embedding-based automated assessment of domain models," in *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems, MODELS Companion 2024, Linz, Austria, September 22-27, 2024*, ACM, 2024, pp. 87–94.
- 10 *Y. Yang, **B. Chen**, *K. Chen, G. Mussbacher, and D. Varró, "Multi-step iterative automated domain modeling with large language models," in *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems, MODELS Companion 2024, Linz, Austria, September 22-27, 2024*, ACM, 2024, pp. 587–595.
- 11 **B. Chen**, *K. Chen, S. Hassani, *et al.*, "On the use of GPT-4 for creating goal models: An exploratory study," in *31st IEEE International Requirements Engineering Conference, RE 2023 - Workshops, Hannover, Germany, September 4-5, 2023*, IEEE, 2023, pp. 262–271.
- 12 **B. Chen**, F. Yi, and D. Varró, "Prompting or fine-tuning? A comparative study of large language models for taxonomy construction," in *ACM/IEEE International Conference on Model Driven Engineering Languages and Systems, MODELS 2023 Companion, Västerås, Sweden, October 1-6, 2023*, IEEE, 2023, pp. 588–596.
- 13 *K. Chen, *Y. Yang, **B. Chen**, J. A. H. López, G. Mussbacher, and D. Varró, "Automated domain modeling with large language models: A comparative study," in *26th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems, MODELS 2023, Västerås, Sweden, October 1-6, 2023*, IEEE, 2023, pp. 162–172.
- 14 **B. Chen**, *K. Chen, *Y. Yang, *et al.*, "Towards improving the explainability of text-based information retrieval with knowledge graphs," in *The First Workshop on Trustworthy Learning on Graphs (TrustLOG-CIKM 2022)*, Atlanta, Georgia, USA, October 2022, 2022.

- 15 **B. Chen**, K. Marussy, S. Pilarski, O. Semeráth, and D. Varró, “Consistent scene graph generation by constraint optimization,” in *37th IEEE/ACM International Conference on Automated Software Engineering, ASE 2022, Rochester, MI, USA, October 10-14, 2022*, ACM, 2022, 25:1–25:13.
- 16 **B. Chen**, D. Havelock, C. Plante, M. Sukkarieh, O. Semeráth, and D. Varró, “Automated video game world map synthesis by model-based techniques,” in *MODELS ’20: ACM/IEEE 23rd International Conference on Model Driven Engineering Languages and Systems, Virtual Event, Canada, 18-23 October, 2020, Companion Proceedings*, ACM, 2020, 4:1–4:5.

Preprints

- 1 M. Saad, J. A. H. López, **B. Chen**, N. A. Ernst, D. Varró, and T. Sharma, “SENAI: towards software engineering native generative artificial intelligence,” *arXiv preprint arXiv:2503.15282*, 2025.
- 2 **B. Chen**, K. Marussy, O. Semeráth, G. Mussbacher, and D. Varró, “Certifying robustness of graph convolutional networks for node perturbation with polyhedra abstract interpretation,” *arXiv preprint arXiv:2405.08645*, 2024.

**Students I have (co-)supervised during the project.*